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Original Article Title: *Effective Training Pressure Gates Recursive Knowledge Degradation in LLMs: A Multi-Axis Dose-Response Study*

To: Knowledge-Based Systems Editor (Elsevier)

Re: Response to Reviewers, Revision Round 1

October 4, 2026

Dear Editor and Reviewers,

We sincerely thank you for the thorough and constructive evaluation of our manuscript. The reviewers identified concrete problems with the original submission: a pipeline inconsistency in the synthetic-exposure experiment, notation ambiguities in the equations, and overclaims about the prospective predictive power of the ETP framework. Addressing them has substantially improved the work. We are grateful for the opportunity to revise and resubmit.

The key enhancements in this revised version include: (1) a corrected and pre-registered dose-response experiment replacing the original C3/C5 comparison, which we identified as being affected by a pipeline inconsistency; (2) a clear separation between ETP-design variables (directly manipulated before training) and ETP-observed post-training diagnostics, with the operational pressure index now labelled as exploratory; (3) seed-level exact permutation tests and paired t -tests with 95 % confidence intervals for all headline comparisons; (4) corrected and fully defined notation for K_0 , $R(t)$, and all symbols in Equations 1–6; and (5) an expanded reproducibility section specifying checkpoint identifiers, decoding configurations, and all scripts required to regenerate tables and figures.

We are submitting: (a) this point-by-point response document; (b) a highlighted revised manuscript showing all changes in blue; (c) a clean final manuscript (LaTeX source + PDF); and (d) the replication package archived at <https://doi.org/10.5281/zenodo.23146342>.

Revised text quoted from the manuscript in this letter appears in [blue](#).

Respectfully submitted on behalf of all authors,

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Editor

Revise (including language editing). Provide point-by-point response to each reviewer comment in the cover letter.

The manuscript has been revised throughout for clarity. The point-by-point response follows.

1 Reviewer 1

R1.1 — Prospective calibration of effective training pressure

How can effective training pressure be quantitatively calibrated and prospectively tested on an unseen backbone or dataset?

The present experiments do not establish a transferable scalar ETP predictor or a prospective threshold for an unseen backbone or dataset. The revised manuscript distinguishes controllable design variables (nominal adapter rank, learning rate, planned training steps, and synthetic-data dose) from post-training diagnostics, including effective rank and weight drift. Under the tested protocol, we identify empirical intervals between adjacent configurations exhibiting different retention trajectories; these intervals should not be interpreted as universal numerical thresholds.

A prospective test would require specifying a calibration rule and outcome criterion before examining a held-out backbone or dataset, selecting configurations on both sides of the predicted regime boundary, and evaluating them without refitting that rule to the held-out outcomes. We have added this as a concrete validation protocol in Section 6 and state explicitly that such held-out prospective validation has not been performed in the present study. Accordingly, we describe effective training pressure as an *operational experimental framework*, not a validated universal predictor.

The pre-registered G5 experiment (three single-seed cells on Qwen) provides limited exploratory evidence: retention was ordered in the expected direction for 2 of 3 pre-specified configurations (the degradative cell was classified as bounded rather than degradative). No operational Π threshold was reported for G5 because the threshold was not pre-specified, and the experiment does not test an unseen backbone or dataset. Prospective calibration remains future work.

Location in revised manuscript: Section 3.5 (ETP as framework, revised); Section 6 (prospective calibration protocol added); Section 4.4 (G5 prospective results).

R1.2 — Separating confounds from pressure effects

How are quantization, optimizer state, data order, and model-specific dynamics separated from the claimed pressure effects?

The revised manuscript addresses each potential confound through a dedicated control, now documented in Section 3.9 with Table 2 (tab:controls).

Regarding quantization: all QLoRA experiments use 4-bit NormalFloat (NF4) quantization with adapters trained in bf16. Ablation G3 reruns the same protocol with standard bf16 precision and no base-weight quantization, three seeds each at $r = 16$ and $r = 256$. At $r = 16$, bf16 achieves Gen10 = 97.4% (76/78 per seed), while the original NF4 single-seed result was 96.2% (75/78); these use different seeds and pipelines and are not a paired comparison. At $r = 256$, bf16 achieves Gen10 = 79.1%, within 0.0 pp of the NF4 G1+G4 mean under the same seeds and matched protocol. Quantization is therefore not the primary driver of the observed regime separation.

Regarding optimizer state: a fresh optimizer is instantiated at every generation; no momentum or adaptive-rate state is carried forward between generations. This is held constant across all conditions within a backbone.

Regarding data order: ablation G4 runs Qwen at $r = 256$ with three independent shuffle seeds (42/101/202), keeping all other conditions identical to G1 (seeds 15/137/256). A permutation test comparing Gen10 retention between the two seed groups gives $p = 0.70$ (two-sided, $N = 3 + 3$). Because seed and data order varied simultaneously in G4, this result does not isolate the effect of order alone; it establishes that the combined variance is statistically indistinguishable from the seed-to-seed variance already present within G1. The combined $N = 6$ seed set (G1+G4) yields Gen10 = $79.7\% \pm 1.7$ pp.

Regarding ETP-design versus ETP-observed: we now separate two classes of ETP variables. *ETP-design* variables are known before training (nominal rank, learning rate, training steps,

and synthetic exposure fraction) and are directly manipulated. *ETP-observed* variables are measured after training (effective rank, mean absolute weight drift $d(t)$, response length, and SDI-3) and serve as post-hoc diagnostic proxies, not pre-specified predictors.

For the central Qwen result, the $r = 16$ condition is represented by three G3/bf16 runs (seeds 15/137/256, Gen10=76/78 per seed) and $r = 256$ by six G1+G4 runs (Gen10=61, 63, 61, 64, 61, 63 out of 78; mean $79.7\% \pm 1.7$ pp). Seeds 15, 137, and 256 appear in both groups, which differ simultaneously in rank, quantization, and data-order protocol; the 17.7 pp arithmetic gap is therefore reported as a descriptive observation, not a confirmatory permutation test. The cleanest rank-only comparison is the G3 within-protocol design (same seeds, same bf16, rank the only variable): per-seed Gen10 differences are +15, +13, and +15 items out of 78, mean +18.4 pp; sign-flip $p = 0.25$ ($N = 3$, all positive direction but underpowered).

For Gemma 3, the $r = 4$ versus $r = 16$ Gen10 contrast ($N = 5 + 5$) gives $\Delta = +37.8$ pp (from raw counts, $|K_0| = 46$). If the same five seeds were used at both ranks (paired design), the exact sign-flip test gives $p = 0.0625$ two-sided; if seeds are independent, a bilateral permutation test over $\binom{10}{5} = 252$ allocations gives $p = 0.0079$. The seed assignment across ranks is not confirmed in the archived data; we report both tests.

Location in revised manuscript: Section 3.5 (ETP definition); Table 1 (tab:notation); Section 3.9 (tab:controls); Section 4.2 (Qwen, G3 paired contrast, 17.7pp descriptive gap); Section 4.3.1 (Gemma 3, $p = 0.0625$ sign-flip or $p = 0.0079$ unpaired permutation); Section 4.6 (G3 and G4 controls).

R1.3 — External validity of the evaluation

Do the conclusions hold for longer answers, multi-hop questions, other domains, and larger factual evaluation sets?

Three distinct concerns are raised: the evaluation metric, the dataset scope, and the extension to other domains.

On verbosity bias: the primary metric uses a bidirectional substring criterion, intentionally permissive to handle short-answer variation and aliasing. Configurations that generate longer responses may receive inflated retention scores. To address this, we performed a strict exact-match re-evaluation on the saved predictions for all reported conditions. Under the strict scorer, absolute retention values are lower, but the direction of every regime classification is unchanged. This re-evaluation is reported in supplementary Table S2 and discussed in Section 6.

On dataset scope: TriviaQA was chosen to isolate factual retention from reasoning or stylistic confounds. Short answers minimise format-dependent evaluation noise, and the substring criterion is resilient to verbosity changes. The limitation is that findings may not transfer to long-form generation, multi-hop reasoning, open-ended dialogue, or code synthesis; this scope restriction is stated explicitly in Section 6.

On evaluation set size: the held-out set contains 200 questions, of which K_0 is the set answered correctly at generation 0 (e.g., $|K_0| = 78$ for Qwen). This set is disjoint from the 2,000-question synthetic training stream. Scaling the evaluation set would alter the training-data composition and is deferred to future work.

Location in revised manuscript: Section 6, ll. 2094–2100; supplementary Table S2; Section 3.1 (evaluation protocol).

R1.4 — Exposure reduction: correction, pre-registered result, and mechanism

Why does the five-percent exposure reduction stabilize Qwen but not Gemma 3, and does this generalize across ranks and seeds?

This question led directly to identifying a pipeline inconsistency in the original submission and to a substantially revised, pre-registered experiment.

The original submission reported that a $\sim 5\%$ reduction of synthetic training examples raised retention by +9.0 and +9.4 pp for conditions C3 and C5 respectively. We subsequently identified that C1 (the baseline) and C5 (the intervention) had been produced by different evaluation pipelines, differing in the K_0 construction procedure and the data-loading code path. When C1 and C5 are re-run under an identical pipeline with paired seeds, the effect of $\sim 5\%$ example removal is only +1.9 pp (not distinguishable from zero at $n = 3$). This discrepancy is acknowledged explicitly in the revised manuscript, and the original C2–C5 results are archived in the replication repository for transparency. This error was uncovered as a direct result of the reviewer’s question.

To replace the discredited result, we designed and pre-registered a dose–response experiment (protocol committed as [a916ebf](#) before any run), varying synthetic exposure at 0%, 10%, 25%, and 50% removal across five paired seeds on Qwen 2.5 1.5B at $r = 256$. At Gen 5, the gains are +3.6 pp (10%, $p = 0.009$), +3.8 pp (25%, $p = 0.003$), and +4.4 pp (50%, $p < 0.001$), $n = 5$ seeds each. The Gen 5 benefit is uniform across doses while the Gen 10 benefit is dose-graded: +1.8 pp at 10% ($p = 0.160$, not significant), +4.9 pp at 25% (95 % CI [2.5, 7.2], $p = 0.005$), and +9.2 pp at 50% (95 % CI [6.9, 11.6], $p < 0.001$). At 50% removal the Gen 5→10 slope is -0.15 pp/generation versus -1.13 pp/generation at 0% (mean per-seed difference of Gen10–Gen5 counts divided by five generations), strongly attenuating progressive loss and shifting the system toward the Bounded regime.

The pre-registered prediction that 50% removal would match halving the learning rate from $\eta = 10^{-5}$ to 5×10^{-6} yielded a similar point estimate: Gen 5 retention at dose = 50% is 89.7% versus 90.2% at $\eta = 5 \times 10^{-6}$ (mean over three seeds; Gen 10 difference +0.4 pp, 90 % CI [−0.8, 1.6]). Formal equivalence was not established.

A parallel pre-registered dose–response experiment on Gemma 3 1B IT at $r = 10$ (G2, three paired seeds, $|K_{0,G2}| = 44$) shows that a 50% reduction raises Gen 10 retention by +12.9 pp (95 % CI [9.70, 16.10], $p = 0.001$). However, the Gen 5→10 slope in the G2 50% arm (-0.91 pp/generation) equals that of the 0% arm (-0.91 pp/generation) within rounding, indicating that the exposure effect on Gemma 3 manifested as a Gen 10 retention gain rather than as a reduction in the rate of progressive loss. The mechanism was not established; a steps-matched control was inconclusive at $n = 3$.

Location in revised manuscript: Section 4.5 (B7 Qwen and G2 Gemma 3 dose–response; pre-registration; LR equivalence); Section 6 (mechanism limitation); replication repository (original C1–C5 results archived).

R1.5 — Reproducibility package

What complete code, checkpoints, prompts, seeds, and raw outputs will support independent reproduction?

Independent reproduction requires more than aggregate retention tables. The reproducibility section (Section A.1) has been expanded to specify the exact model checkpoint identifiers for all three backbones; train and evaluation question IDs and the baseline-correct K_0 sets ($|K_0| = 78$ for Qwen; two Gemma 3 sets: $|K_{0,\text{rank}}| = 46$ for rank-sweep experiments and $|K_{0,G2}| = 44$ for G2; see Section 3.6); prompt and chat templates for both synthetic data generation and deterministic

evaluation; distinct decoding settings (`do_sample=False` and `temperature=0` for evaluation); correctness normalization and alias-expansion code; training configurations, software versions, random seeds, data-order and downsampling masks; per-generation retention counts and boolean correctness vectors included in the consolidated CSV; and scripts to regenerate every table and figure.

The replication package contains all analysis and figure-generation scripts; the consolidated results CSV (G1–G5, 27 executed training runs, per-generation per-seed retention); all 11 publication figures at 300 dpi; and the pre-registration commit for B7 ([a916ebf](https://doi.org/10.5281/zenodo.23146342)). The package does *not* contain raw token-level predictions, adapter checkpoints (available on request; ≈ 70 MB each in bf16), or base-model weights (obtained from their original distributors). It is archived at <https://doi.org/10.5281/zenodo.23146342>. Release v1.0.3 contains the initial replication scripts and data; the revised analysis scripts and updated `numbers.tex` from the current submission are not yet included and will be deposited in a subsequent release. Running `python scripts/make_all.py` reproduces all 112 canonical numbers (15/15 checks pass).

Location in revised manuscript: Section A.1 (repository and data access, expanded).

2 Reviewer 2

R2.1 — Abbreviations and notation

ETP appears in Figure 1 without explicit definition. MTL D is not expanded. KL, JS, and coverage appear in Figure 1 without definition. bf16, SVD, TCE, QLoRA, and LoRA lack consistently expanded forms. “Homeo.” is used in Table 2 without definition. K_0 is used both as a set and as a metric. T , $D_{1,T}$, D_{10} , ΔI , and related symbols in Eq. 6 are not fully defined.

The revised manuscript does not use “ETP” as a standalone acronym; the term is introduced in full at l. 72 and used in full throughout. Figure 1 no longer contains the abbreviation in isolation.

MTLD now reads “[Measure of Textual Lexical Diversity \(MTLD\)](#)” at first occurrence, followed by the functional description.

KL, JS, and coverage were conceptual labels for diagnostics never computed or reported. They have been removed from Figure 1 entirely; the revised caption describes only quantities that are measured: factual retention, effective rank, and distributional shift.

All acronyms now have explicit parenthetical expansions at first use: “[full fine-tuning \(FFT\)](#)” at l. 75, “[singular value decomposition \(SVD\)](#)” at l. 332, “[Quantized Low-Rank Adaptation \(QLoRA\)](#)” at l. 333. TCE (Truncated Cross-Entropy) is cited in the Related Work section as a mitigation method from prior literature; it does not appear in the Methods or Results of this paper. The abbreviation “Homeo.” has been expanded to “[Homeostatic](#)” in all four occurrences in Table 2.

A dedicated notation table (Table 1) distinguishes the two uses of K_0 : K_0 denotes the *set* of held-out questions answered correctly at generation 0; $|K_0|$ denotes its cardinality (e.g., $|K_0| = 78$ for Qwen); $R(t) = |K_t|/|K_0|$ denotes the retention ratio at generation t .

All symbols in Eq. 6 (SDI-3) are now defined in Table 1 and in the inline text preceding the equation: T = evaluation horizon in generations; $D_{1,t}$ = Distinct-1 unigram diversity of synthetic responses at generation t ; ΔI = $\text{Instability}(T) - \text{Instability}(1)$, where $\text{Instability}(t)$ is the fraction of training items whose normalized response differs between generation $t-1$ and generation t ; MeanLen_t = mean response length in words at generation t .

Location in revised manuscript: Abstract l. 72; Figure 1 caption ll. 692–698; Table 1 (tab:notation), ll. 438–445; Equations 1, 3, 5, 6; l. 875 (MTLD); ll. 75, 332, 333 (acronyms); Table 2 (tab:rank_sweep) regime column.

R2.2 — Equations and tables

Equations 1–6 are malformed in the PDF. Eq. 2 for effective rank is missing the negative sign and summation. Eq. 6 for SDI-3 has unclear signs and terms. “ 5×10^6 ” in Table 7 appears to be a rendering error.

The malformed rendering was a font-encoding artefact: Unicode characters in mathematical mode were not mapped correctly to the T1 font metric. The revised submission uses the `lmodern` package with T1 encoding throughout, which eliminates the rendering artefact.

Eq. 1 now reads:

$$R(t) = \frac{|\{q \in K_0 : q \text{ correct at gen. } t\}|}{|K_0|}.$$

Eq. 2 now reads:

$$\text{erank}(BA) = \exp\left(-\sum_i \hat{\sigma}_i \log \hat{\sigma}_i\right),$$

where $\hat{\sigma}_i = \sigma_i / \sum_j \sigma_j$ are the normalised singular values of BA , distinct from the raw singular values σ_i , consistent with Table 1. This is the standard effective-rank definition of [?].

Eq. 6 now reads:

$$\text{SDI-3} = \log \frac{\text{MeanLen}_T}{\text{MeanLen}_0} + \log \frac{\text{D1}_0}{\text{D1}_T} + \Delta I.$$

The first term is positive when mean response length grows; the second is positive when Distinct-1 diversity declines; and ΔI captures item-level instability accumulation. A positive SDI-3 score indicates cumulative distributional deterioration.

The value in Table 7 is the learning rate 5×10^{-6} ; the exponent -6 was dropped in the original PDF by the same encoding issue. The revised PDF renders the value correctly throughout.

Location in revised manuscript: Preamble (lmodern + T1); Eq. 1, l. 595; Eq. 2, ll. 798–803; Eq. 6, ll. 881–886; Table 7 (tab:ranklr), l. 1473.

R2.3 — Technical and data inconsistencies

Qwen threshold: “up to effective rank 150” contradicts Table 2. $r = 128$ called “above-threshold” but classified as Bounded. $r = 256$ called “boundary configuration” but classified as Degradative. Gemma 3: “ $r \geq 16$ ” omits $r \in \{10, 12, 14\}$ already degradative at Gen 5. Abstract overclaim on seed counts. FFT on Gemma 3: non-monotonic result too strongly concluded. C5 removal percentage discrepancy (15% examples vs. 5% tokens).

The phrase “up to effective rank 150” has been removed. All threshold language now uses “an abrupt, threshold-like transition within the tested grid”, reflecting that the boundary is established empirically between adjacent tested configurations, not extrapolated.

The phrase “above-threshold configurations” now refers exclusively to $r = 256$ (Degradative). The $r = 128$ configuration (Bounded) is described as a “transition-zone configuration” where distributional signatures begin to emerge without producing the full degradative trajectory.

The five occurrences of “boundary configuration” referring to $r = 256$ have been replaced throughout with “lowest-pressure degradative configuration tested”, making explicit that the transition itself is localised between $r = 128$ and $r = 256$.

Table 4 reports retention of 78.3%, 73.9%, and 69.6% for $r = 10, 12$, and 14 at Gen 5, all classified as Degradative. The text now reads “the Gemma 3 regime boundary lies between effective ranks 3 and 6, below $r = 10$ ”; the phrase “ $r \geq 16$ ” has been replaced with “ $r \geq 10$ on Gemma 3 (including $r \in \{10, 12, 14\}$ at Gen 5)”.

The abstract now reads “three to five independent seeds for headline conditions (single seed for intermediate ablations)”, accurately reflecting the design. This distinction is also stated in the Methods and reported per table via the added N column.

For FFT on Gemma 3, the word “confirms” has been replaced with “is consistent with” and the non-monotonic pattern (78.3%, 54.3%, 63.0% for $\text{LR} = 10^{-6}, 5 \times 10^{-6}, 10^{-5}$) is acknowledged: the result is treated as supporting evidence at $n = 1$ seed rather than confirmatory.

The 15%/5% discrepancy: C5 removed 15% of training *examples* (questions), which reduced the token count by approximately 5% because responses vary in length. This distinction is now clarified at both locations. The more substantive resolution is that the C1/C5 pipeline inconsistency renders the original comparison uninformative; the dose-response in Section 4.5 (B7) replaces those comparisons.

Location in revised manuscript: Abstract l. 69, l. 72; Contribution #1 l. 95; Section 4.4.3, ll. 1061, 1570; ll. 1737, 1915, 2063, 2069, 2159; Section 5.2; Table 4 (tab:gemma3), ll. 1265, 1277–1280; ll. 1147–1149 (FFT Gemma 3); Section 3.7 (C5 doses); Section 4.5 l. 1708–1713.

R2.4 — Methodology

No formal statistical tests or confidence intervals. ETP framework is not quantified. Cross-architecture normalisation failed with no alternative. ETP is a conceptual label, not a new quantitative measure. Gemma 4 E2B lacks a citation. “Sharp transition” claim weakened by sparse rank grids.

The revised manuscript adds seed-level exact permutation tests for the comparisons where independent training runs are available, treating seeds and runs as the experimental unit. For Qwen, the 17.7pp arithmetic gap between $r = 16$ and $r = 256$ is reported as a descriptive observation because the two groups are not exchangeable under a rank-only null (they differ simultaneously in rank, quantization, and data-order protocol). The cleanest within-protocol comparison is G3 paired design: mean +18.4pp, sign-flip $p = 0.25$ ($N = 3$, all positive direction but underpowered). For Gemma 3, bilateral permutation $p = 0.0079$ (independent seeds) or sign-flip $p = 0.0625$ (paired, if same seeds at both ranks). For the dose-response experiments, paired t -tests with 95% confidence intervals are reported per dose level against the unmodified baseline.

ETP is presented as an *operational experimental framework* rather than a validated scalar predictor. We distinguish ETP-design variables (rank, learning rate, exposure fraction, training steps) from ETP-observed post-training diagnostics (effective rank, $d(t)$, response length, SDI-3). The product $\text{erank} \times \text{LR}$ is an exploratory index labelled as such; it is not used as a prospective predictor.

The failed normalisations in Table 6 do not provide a universal scalar; the table is retained as evidence of incommensurability. Architecture-specific calibration requires a held-out backbone or a pre-specified normalization rule with out-of-sample validation, deferred to future work.

The contribution is the factorial design, the dose-response evidence across three backbones, the

pre-registered exposure result, and the identification of architecture-specific regime boundaries, not the ETP label itself.

The Gemma 4 E2B backbone is now cited as [?] (Gemma Team, 2026, arXiv:2607.02770) at every introduction point. Architecture details are sourced from the released model configuration in the same technical report.

“Sharp transition” has been replaced throughout with “[abrupt, threshold-like transition within the tested grid](#)”, reflecting that the boundary is established empirically between adjacent rank values rather than identified at a precise point.

Location in revised manuscript: Section 4.2 (Qwen, G3 paired contrast); Section 4.3.1 (Gemma 3, $p = 0.0625$ or $p = 0.0079$); Section 4.5 (dose-response, paired t-test); Section 3.5 (ETP design vs. observed); Table 1 (tab:notation); Section 6 (cross-architecture limitation); Section 3.3 (Gemma 4 citation); Abstract l. 72.

R2.5 — Format and citations

References [12] and [33] share arXiv:2408.14572. Reference [32] year inconsistency. Gemma 4 E2B lacks citation. Keisha et al. is a preprint. Reference formatting is inconsistent.

The arXiv number 2408.14572 now appears exactly once, assigned to [?] (CURLoRA, Fawi, 2024). The duplicate entry has been removed.

The entry `ding2024ranktrade` (Rathore et al.) now carries `year = {2025}`, consistent with the December 2025 arXiv submission date; the paper appeared in AACL-IJCNLP 2025.

The Gemma 4 E2B backbone is now cited as [?] at every introduction and description point.

[?] is a non-peer-reviewed preprint (arXiv:2509.04796). This is now declared at first citation: “[?] ([non-peer-reviewed preprint](#)) [report a complementary observation...](#)” The paper is used for contextual framing only; no quantitative claim depends exclusively on it. [?] (published in *Nature*) is added as the primary citation for model-collapse phenomena.

The reference list has been audited. Published works include their full venue with volume, pages, and DOI where available. Works existing only as preprints are formatted as `arXiv preprint arXiv:XXXX.XXXXX`. Entries previously missing venue data (`ding2024ranktrade`, `fawi2024curlora`, `zibakhsh2024`) have been completed.

Location in revised manuscript: References (cas-refs.bib): entries `fawi2024curlora`, `ding2024ranktrade`, `gemma4_2026`, `keisha2025`, all entries (format harmonised).